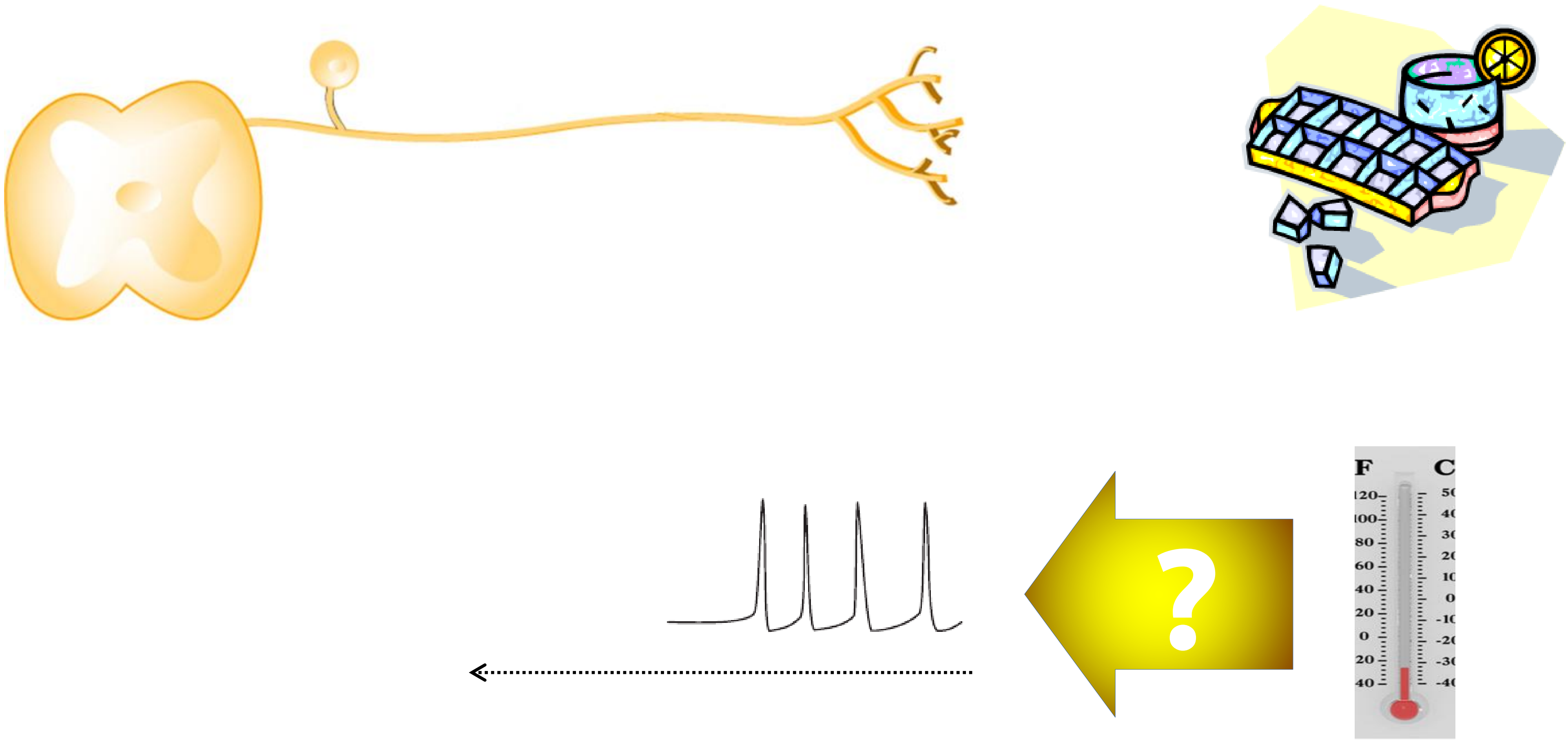
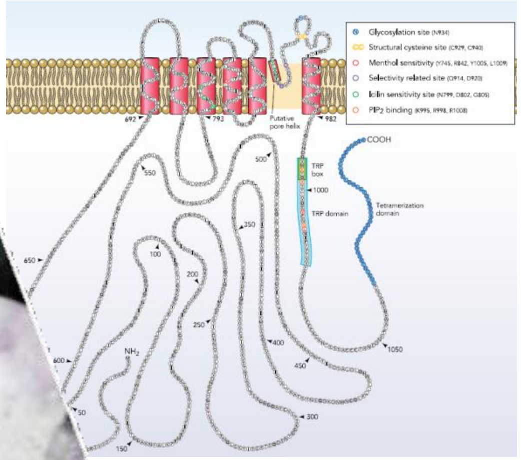
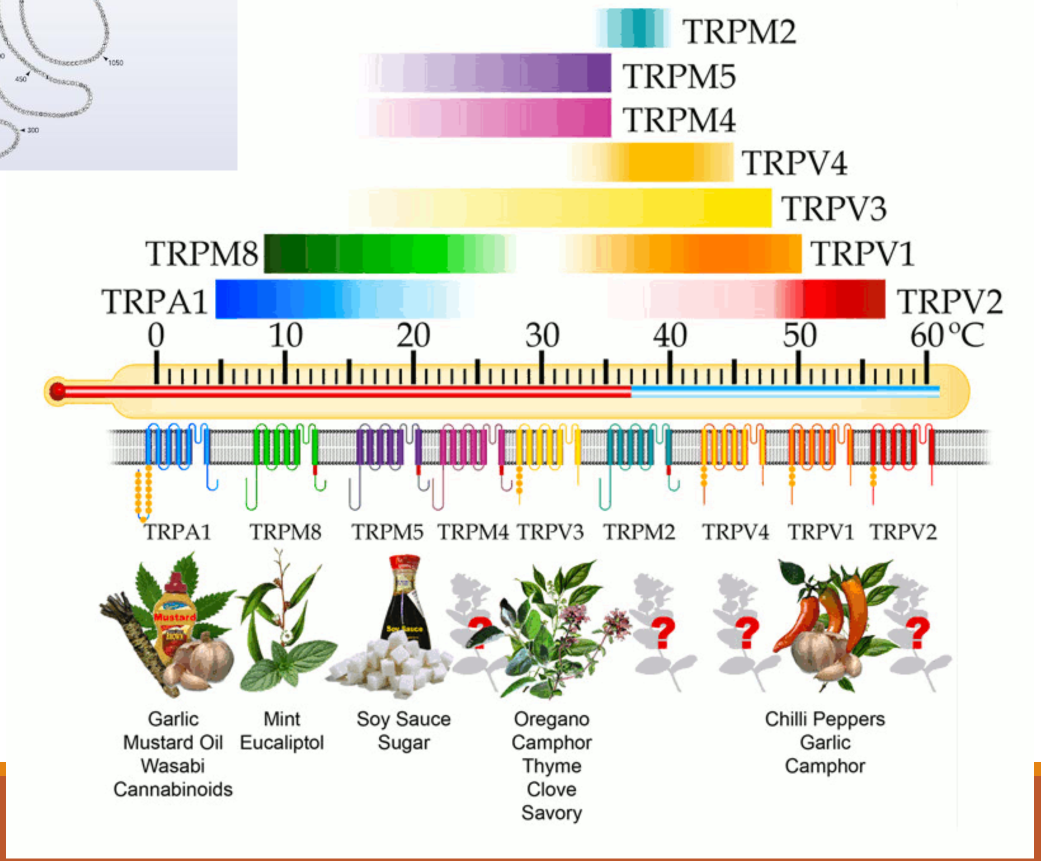
Modelling Thermoreceptors

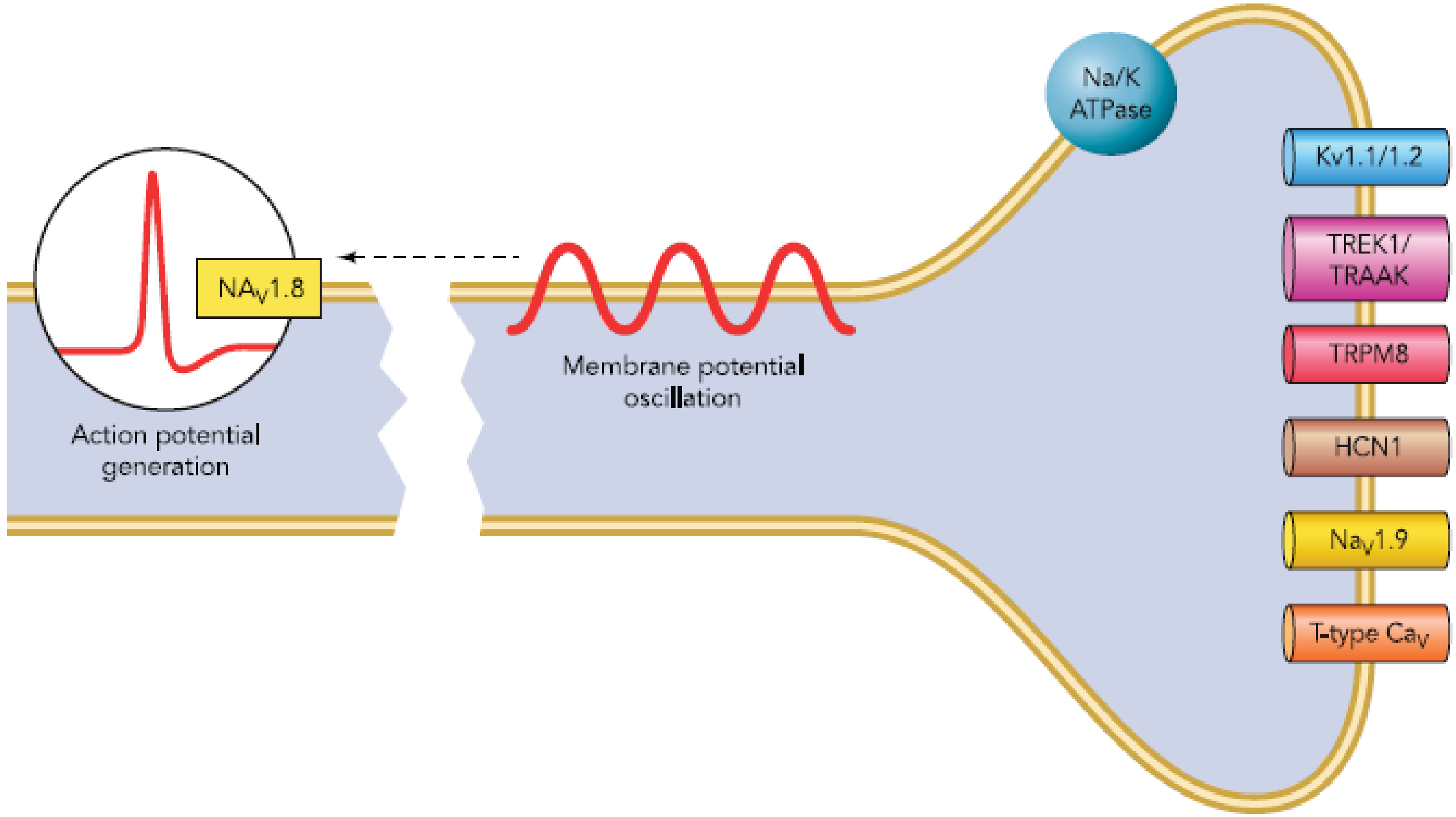
Cold *transduction*



The cold ‘receptor’ TRPM8

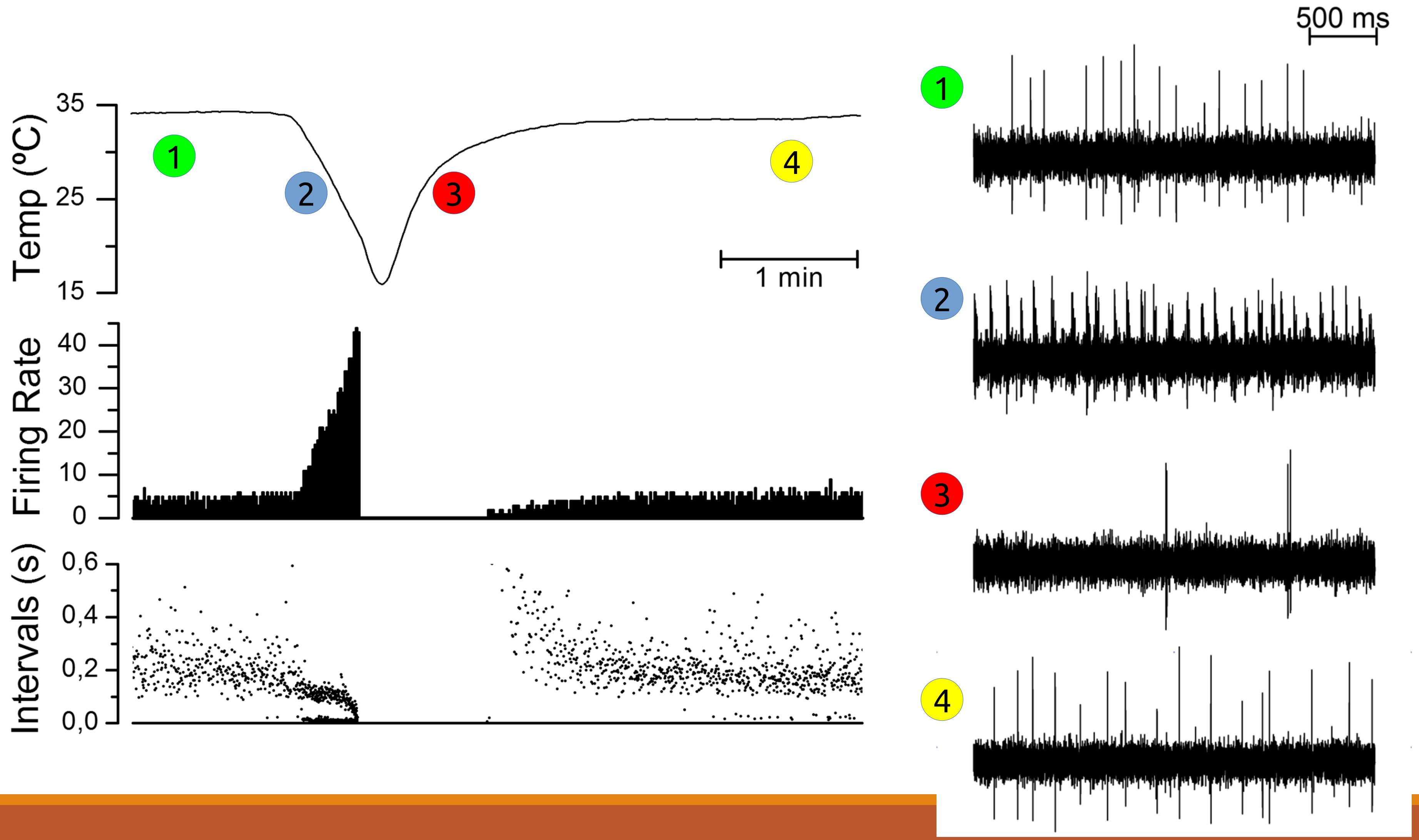






NA/K ATPase: Pierau *et al.* 1975, Schafer&Braun 1990, Spray 1974. **TREK1/TRAAK:** Noel *et al.* 2009, Reid&Flonta 2001, Viana *et al.* 2002. **HCN1:** Orio *et al.* 2009, Orio *et al.* 2012. **K_V1.1/1.2:** Viana *et al.* 2002, Madrid *et al.* 2009. **T-type Ca_V:** Schafer *et al.* 1991. **Na_V1.8:** Zimmermann *et al.* 2007. **Na_V1.9:** Brock *et al.* 2001, Herzog *et al.* 2001

The Dynamic Response to Cold

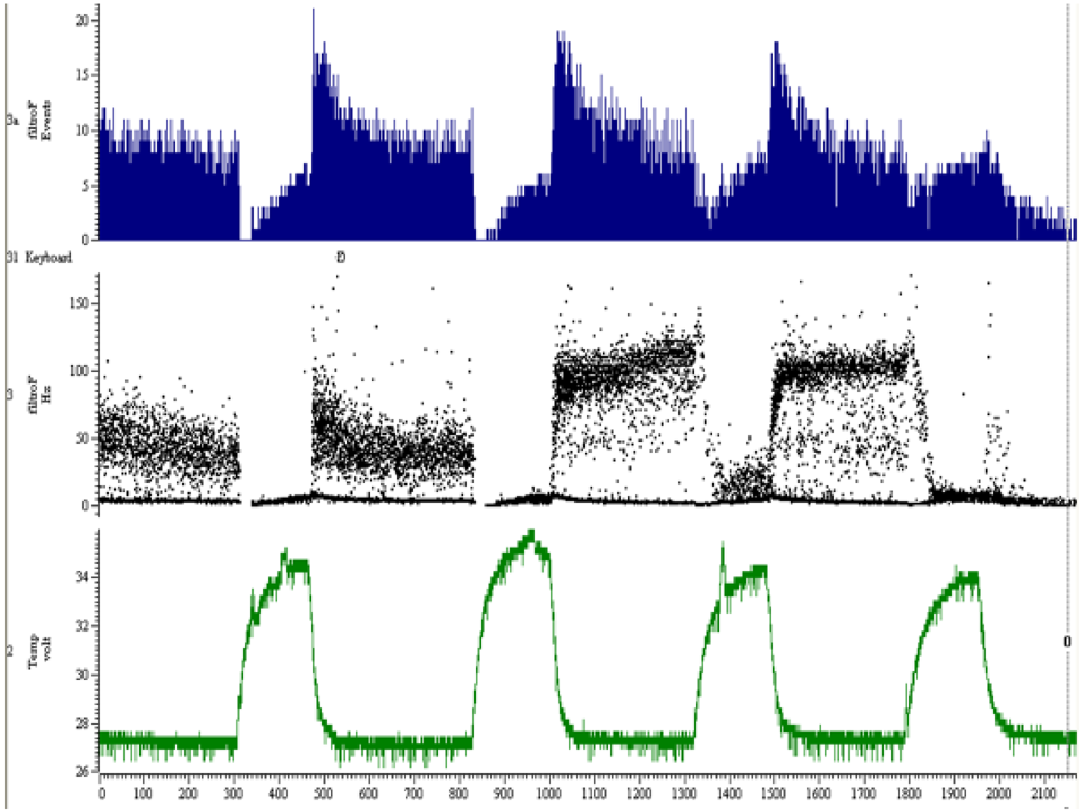


Change Detector

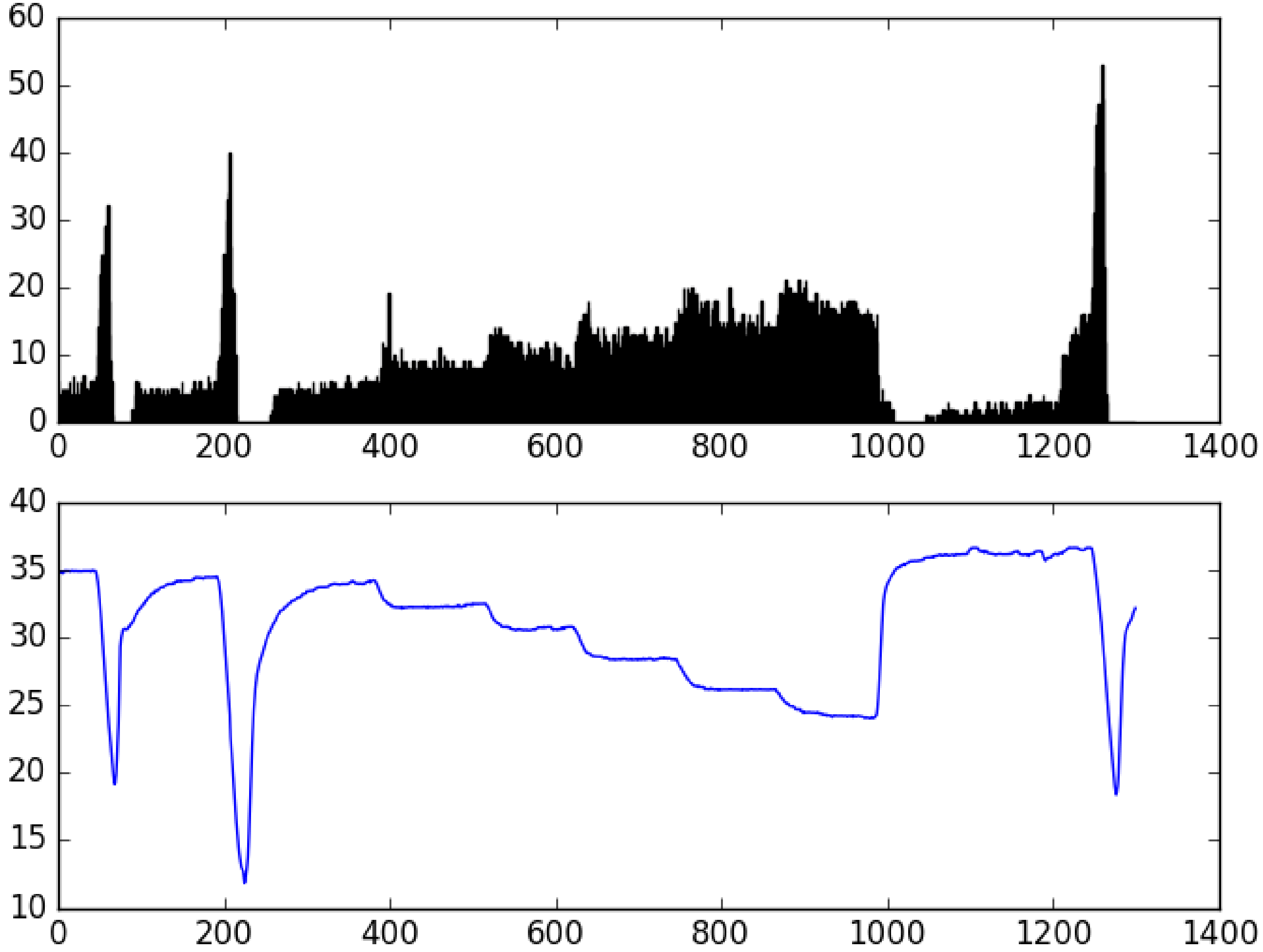
Spikes / seg

Inst. Freq

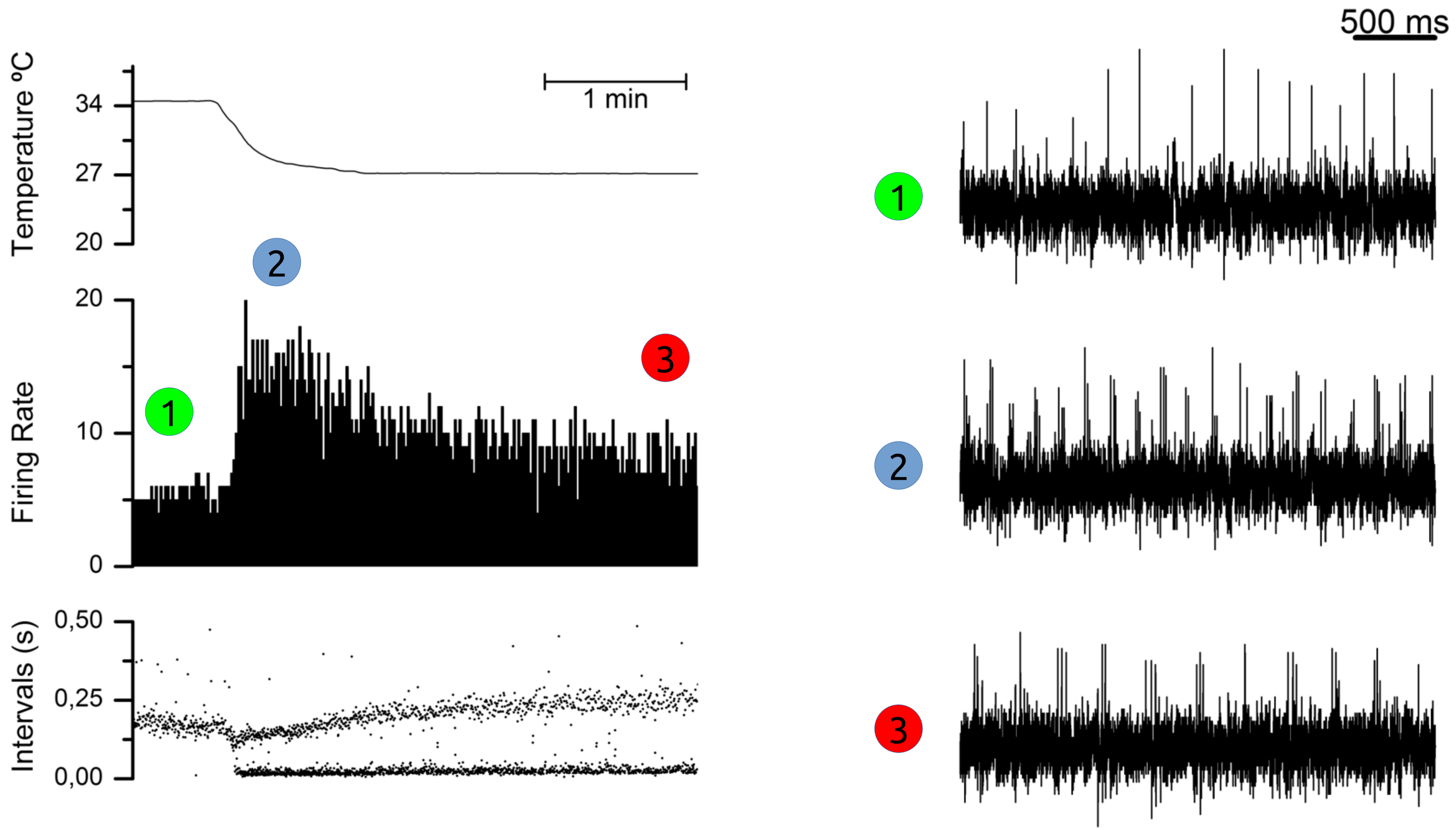
Temperature



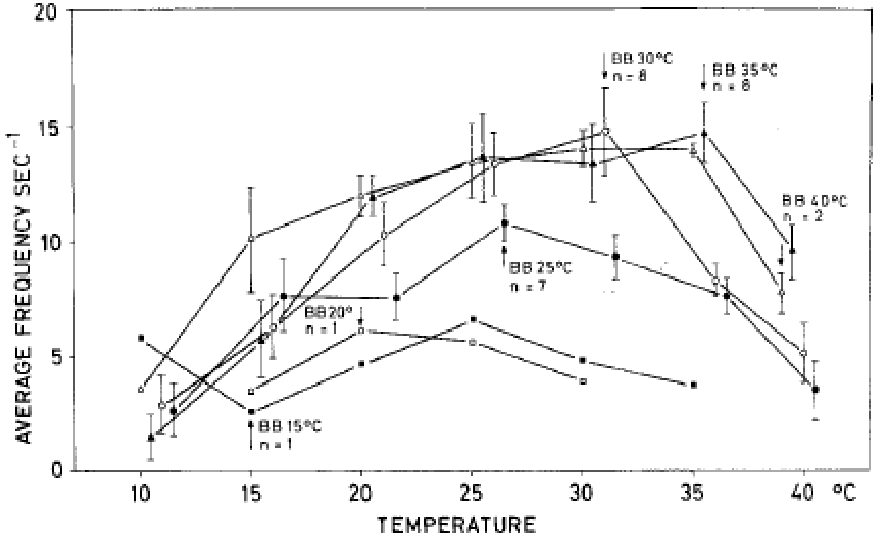
Dynamic response

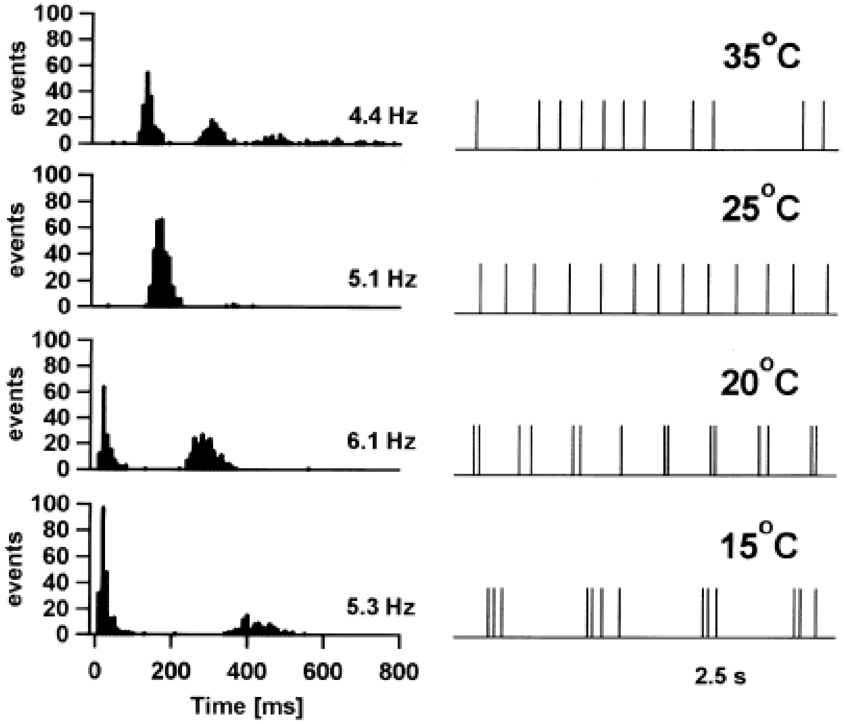


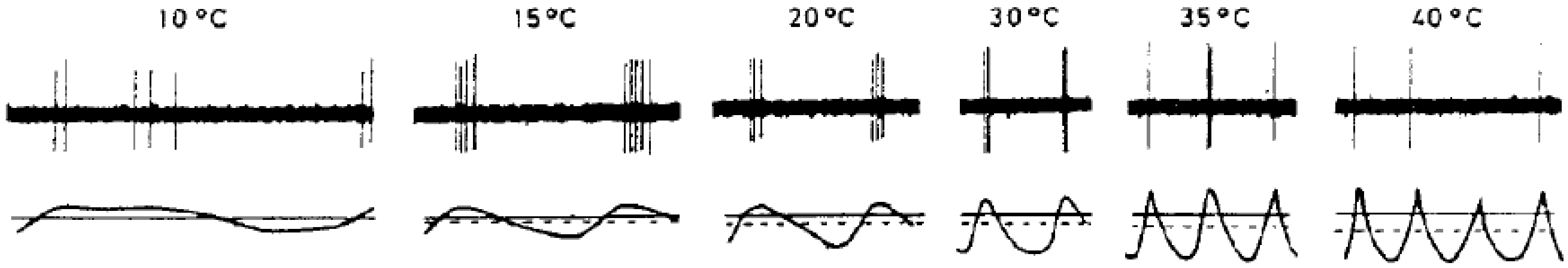
The Static response to temperature



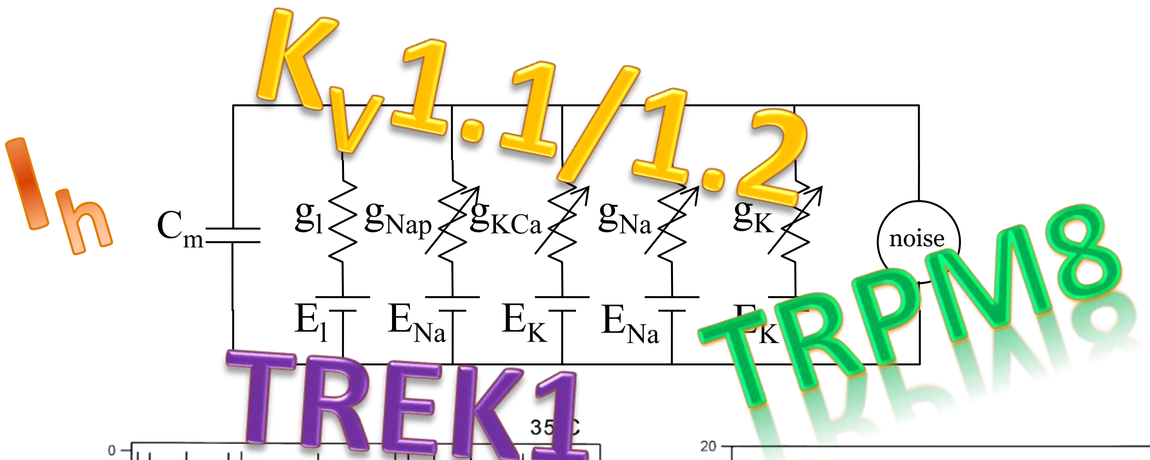
Static Temperature Coding







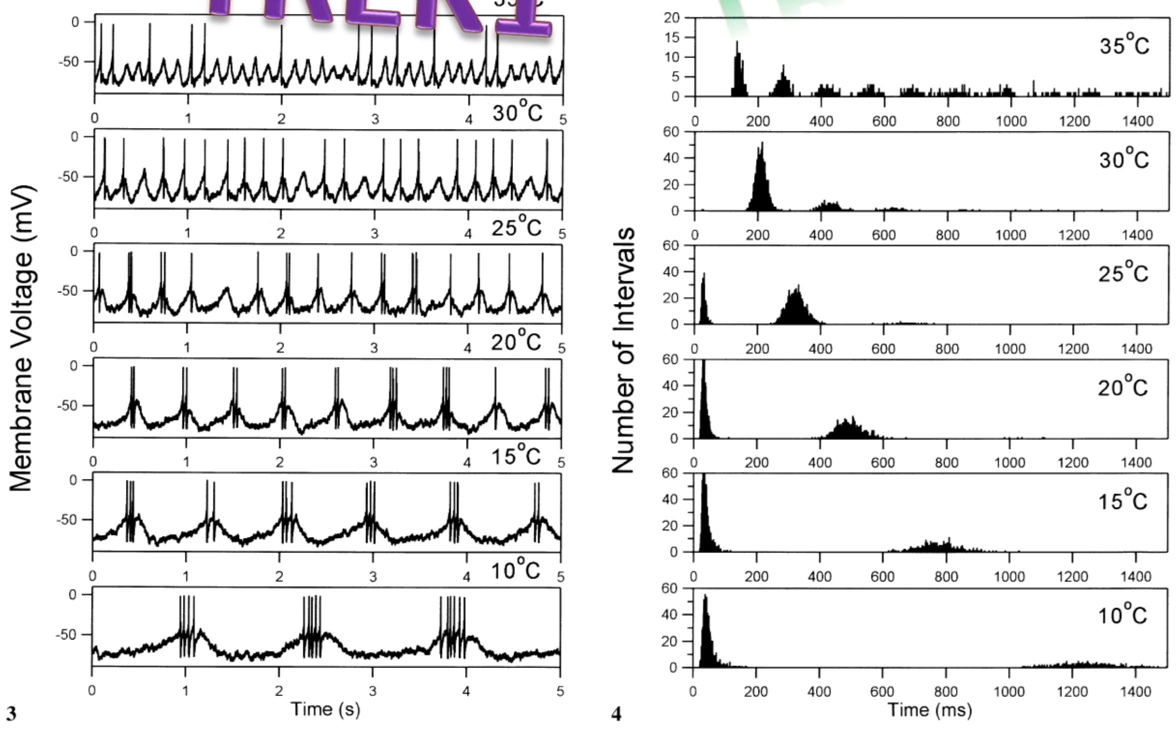
Huber & Braun's model of cold-modulated oscillation



$$C_m \frac{dV}{dt} = -g_{sd}a_{sd}(V - E_{Na}) - g_{sr}a_{sr}(V - E_K) -$$

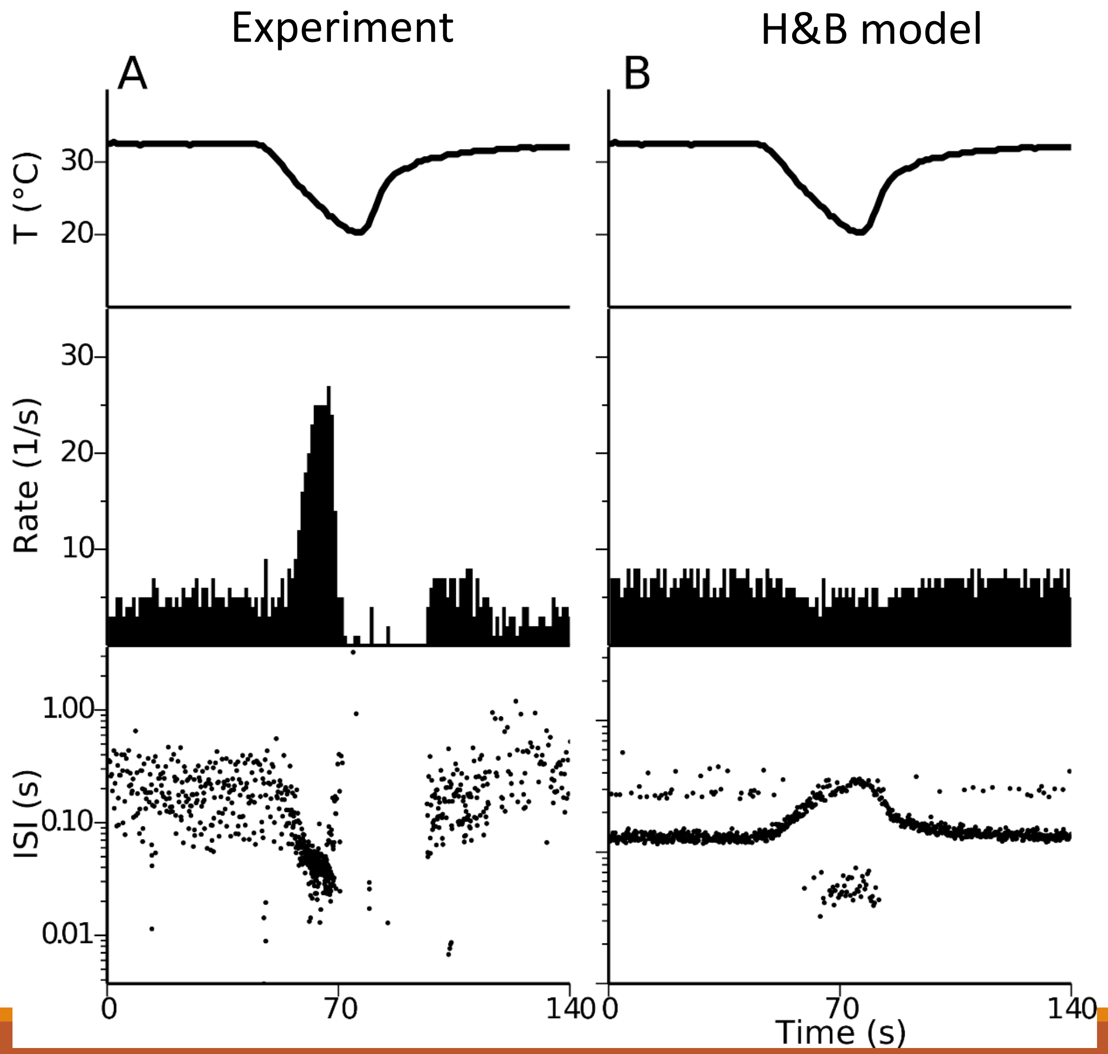
$$\frac{da_{sd}}{dt} = \varphi_T \frac{a_{sd}^{\infty} - a_{sd}}{\tau_{sd}}$$

$$\frac{da_{sr}}{dt} = \varphi_T \frac{\eta I_{sd} - \kappa a_{sr}}{\tau_{sr}}$$



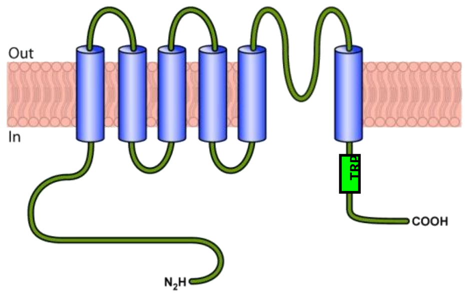
Braun, H.A. *et al* (1998) Int. J. Bif. Chaos
Huber, M.T. *et al* (2000) Chaos Sol. Fract.

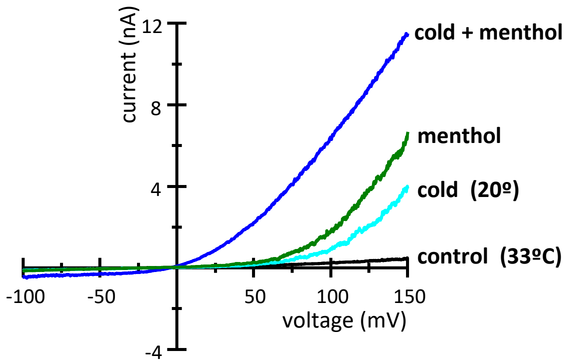
H&B model lacks dynamic response



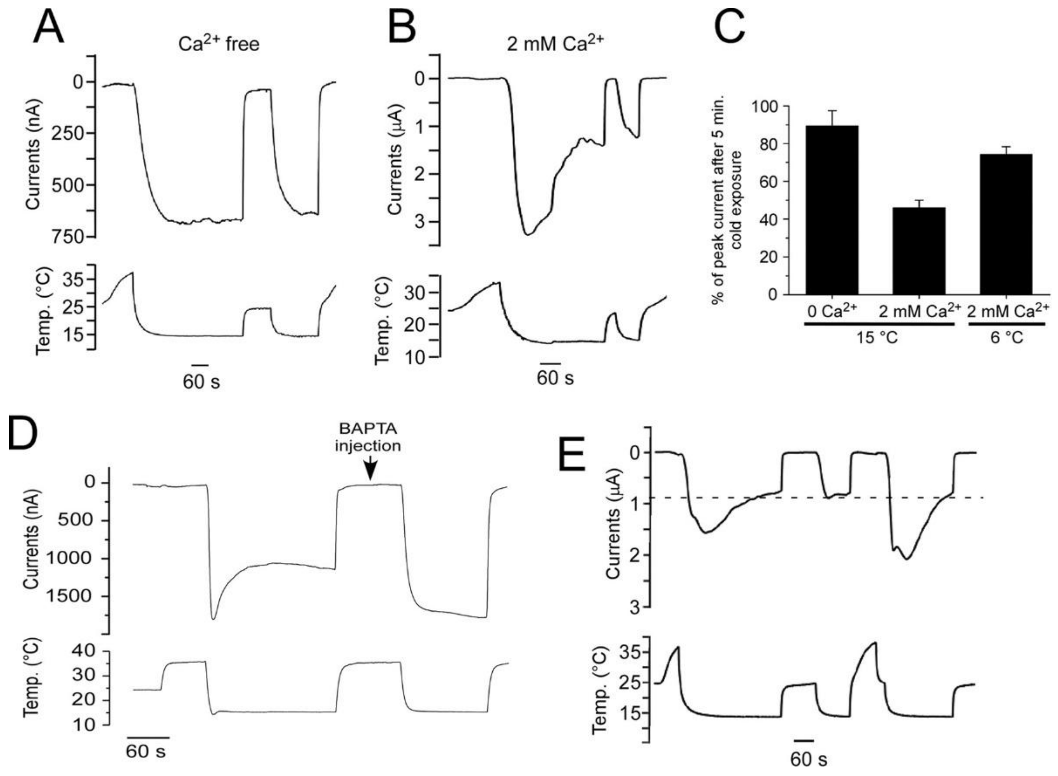
The obvious suspect: TRPM8 channel

TRPM8



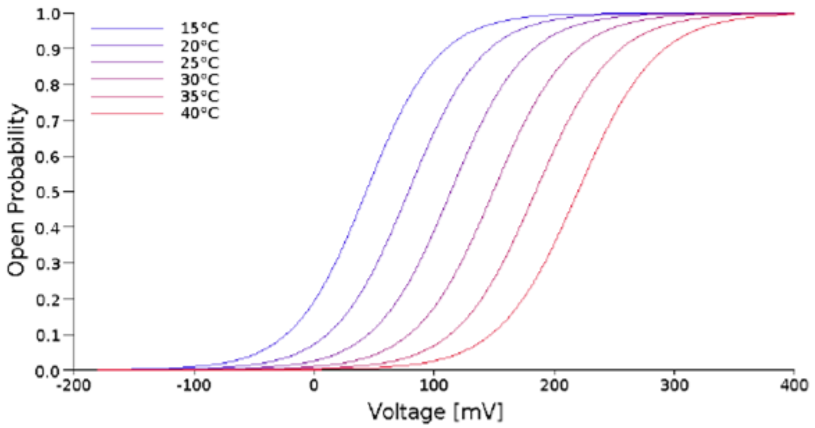


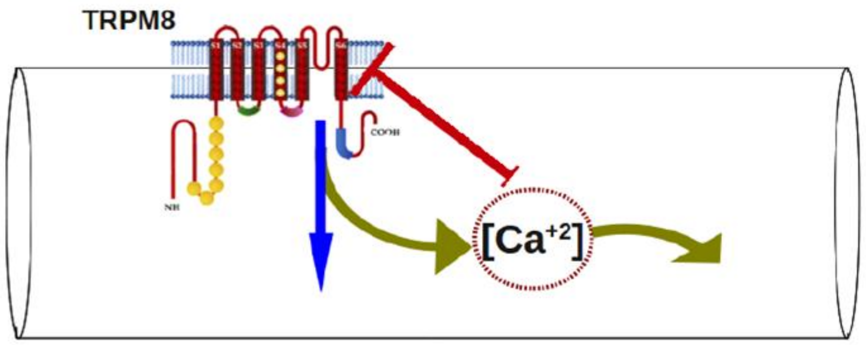
Ca^{2+}-dependent adaptation of TRPM8 current



Daniels, Takashima, McKemy (2009) *J Biol Chem*

TRPM8 model





$$I_{M8} = \bar{g}_{M8} a_{M8} (V - E_{M8})$$

$$a_{M8} = \frac{1}{1 + \exp\left[-\frac{zF}{RT}(V - V½)\right]}$$

$$V_{½} = V_{½}^{0}(T) + \Delta V$$

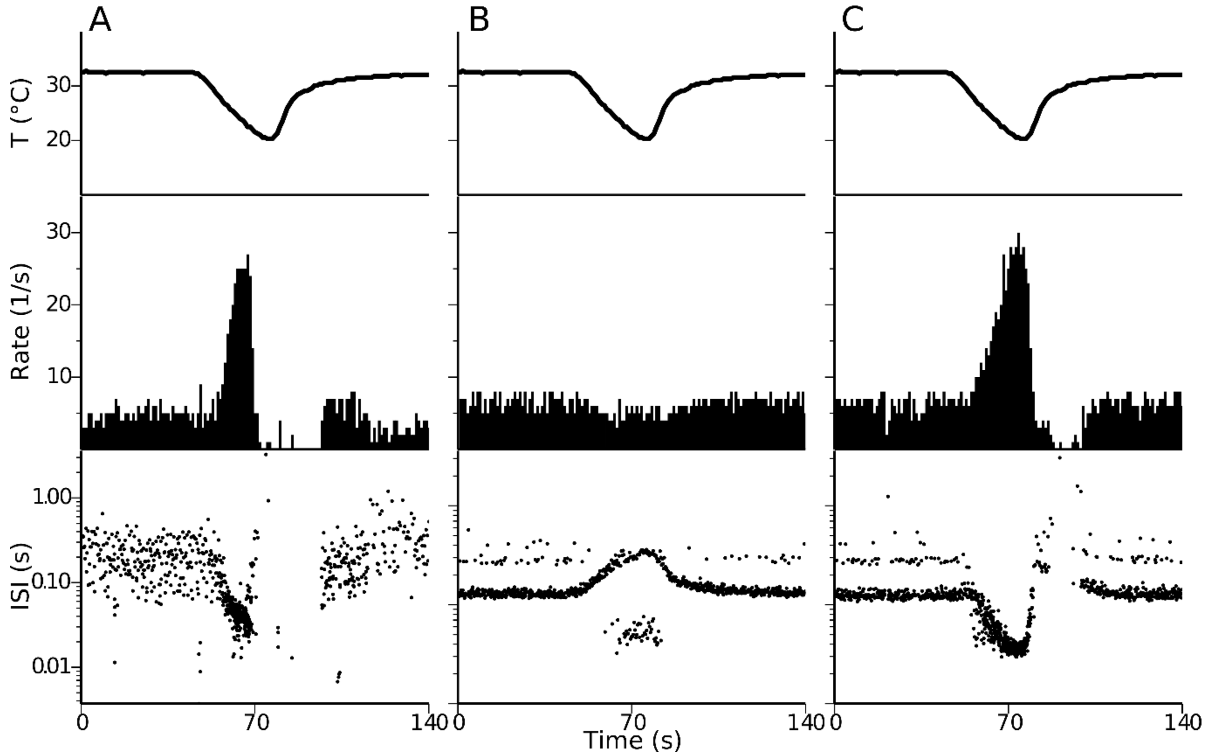
$$\Delta V_{\infty} = \Delta V_{min} + (\Delta V_{max} - \Delta V_{min}) \frac{[Ca^{2+}]}{[Ca^{2+}] + K_d}$$

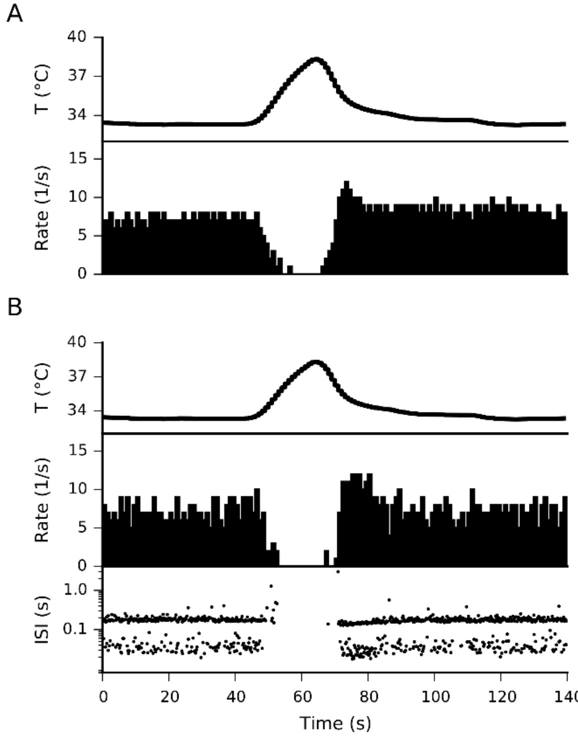
$$\frac{d[Ca^{2+}]}{dt} = -\frac{p_{Ca} I_{M8}}{2Fd} - \frac{[Ca^{2+}]}{\tau_{Ca}}$$

$$\frac{d\Delta V}{dt} = \frac{\Delta V_{\infty}([Ca^{2+}]) - \Delta V}{\tau_{\Delta V}}$$

TRPM8 model produces dynamic response

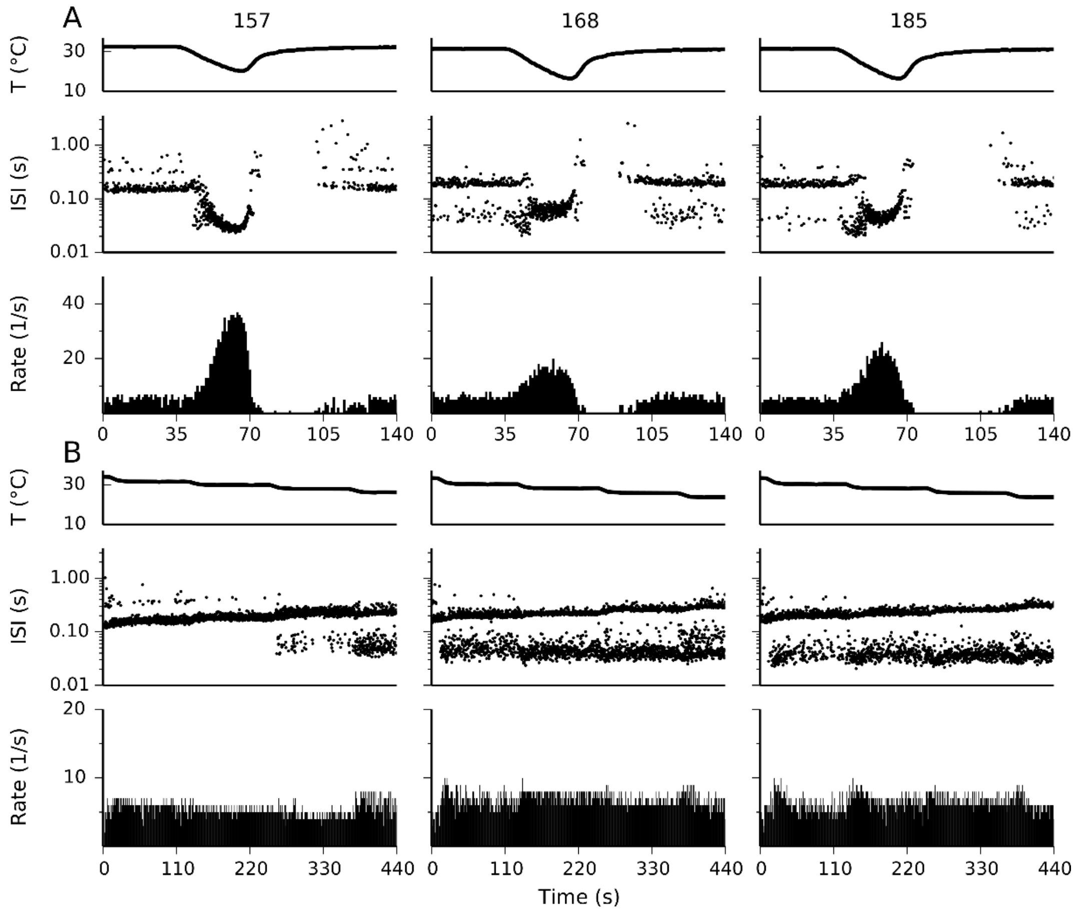
Experiment H&B model H&B + M8

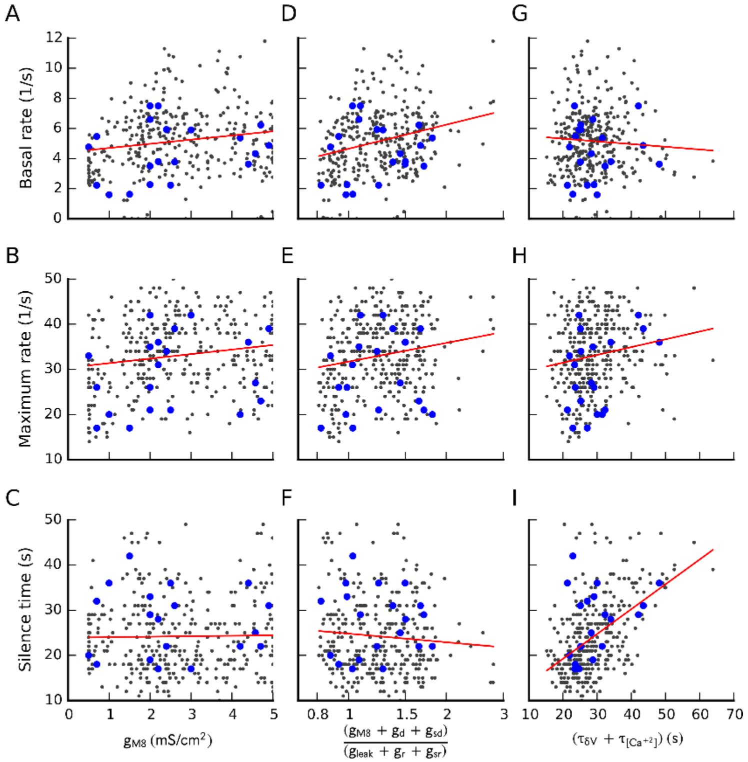




Olivares, E. et al, *PLOS One* 2015

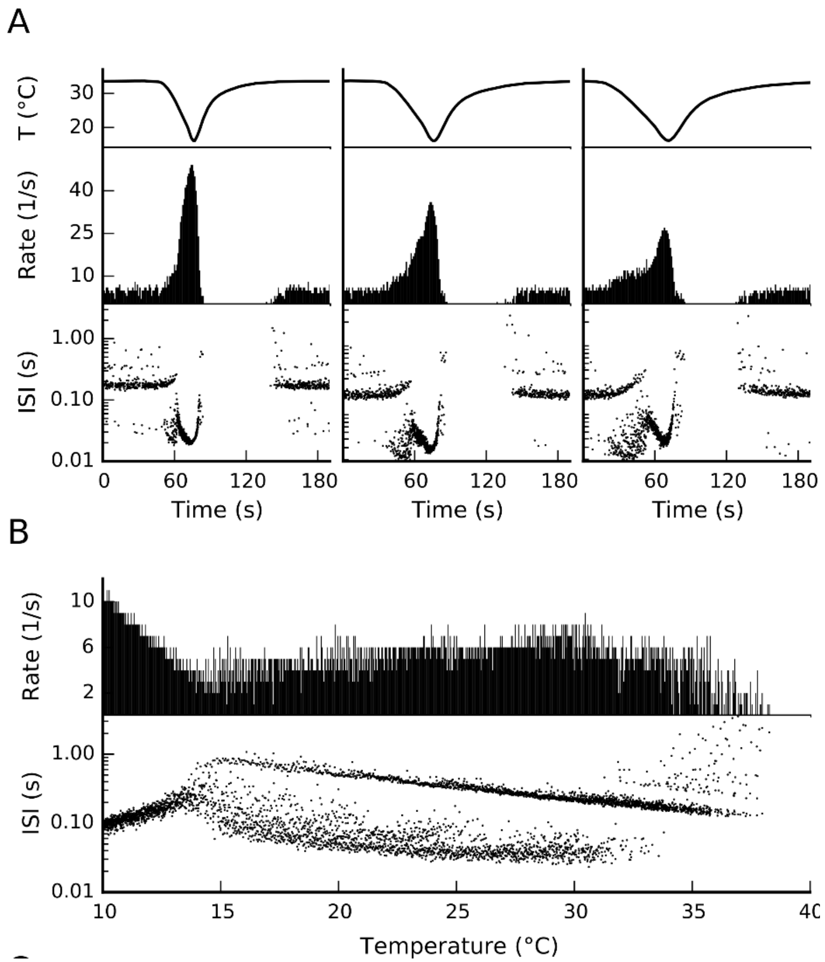
Biological variability reproduced

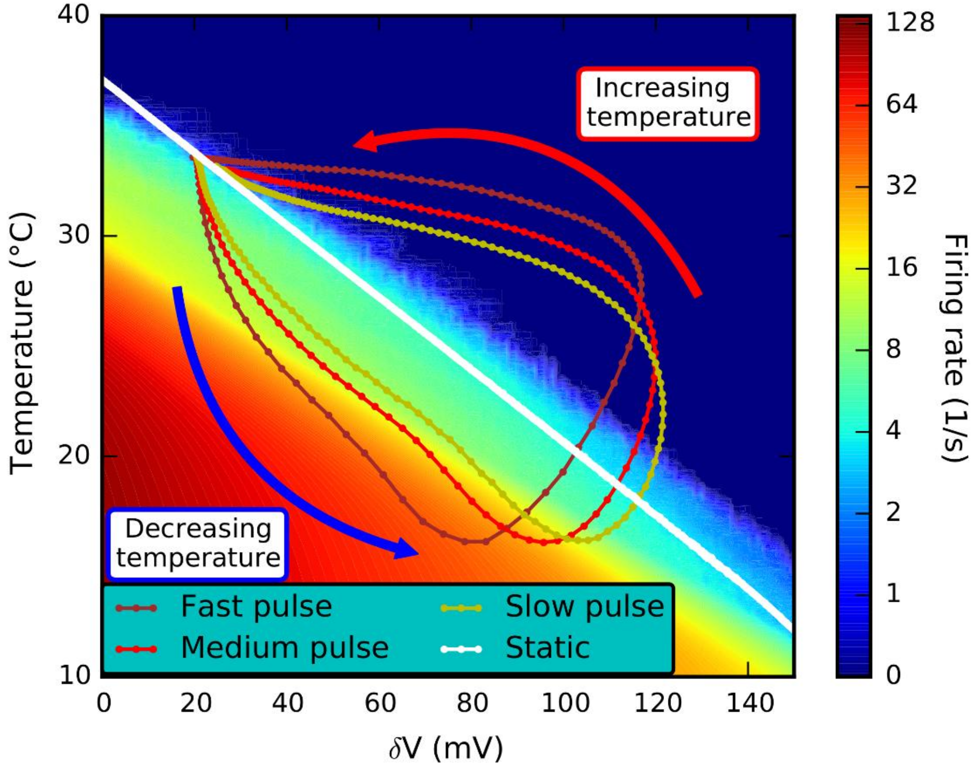




Olivares, E. et al, *PLOS One* 2015

Understanding hysteresis

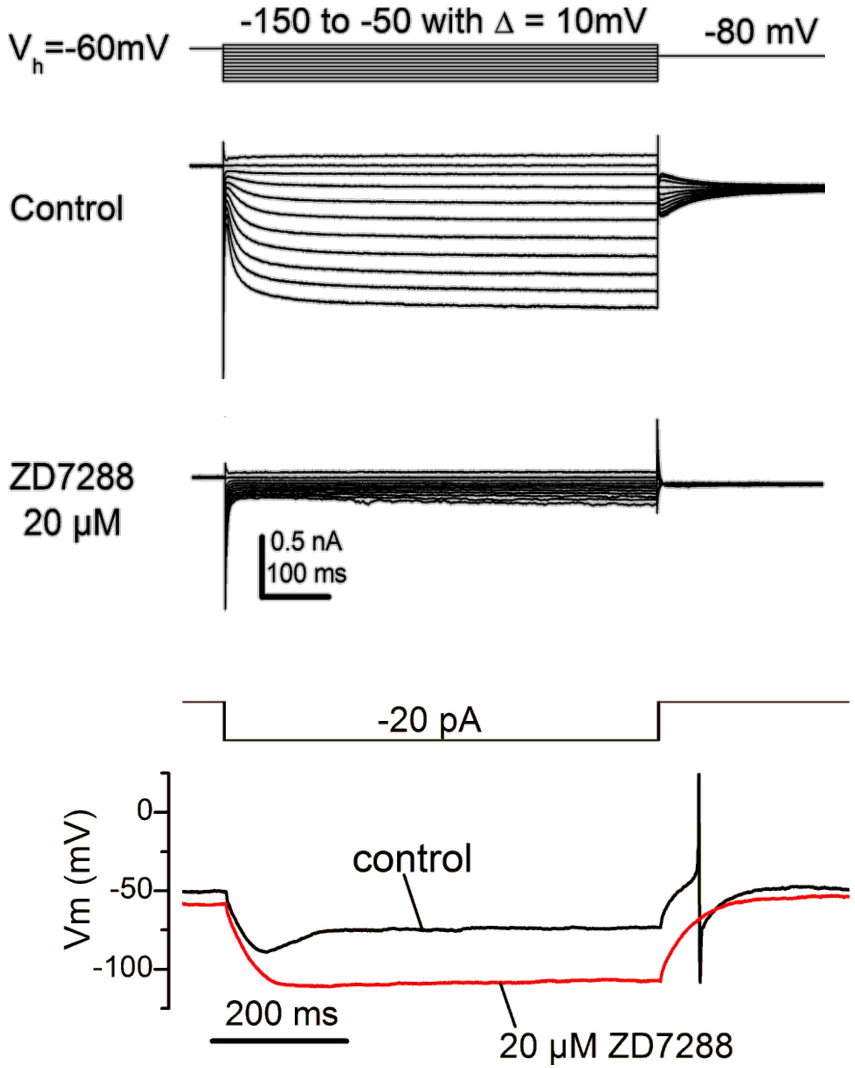


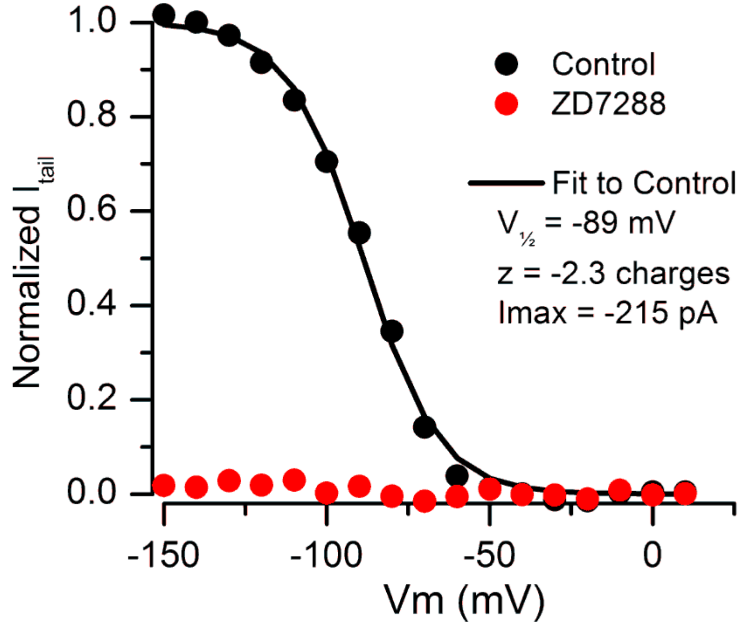


(*degree of adaptation*)

Olivares, E. et al, *PLOS One* 2015

I_h in cold-sensitive (CS) neurons



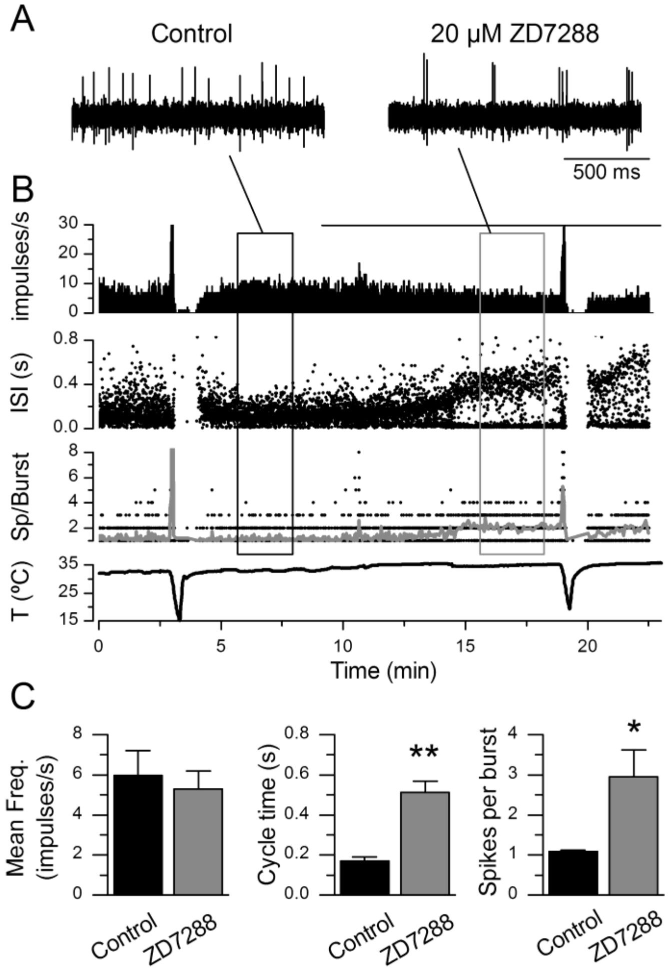


Orio et al., *J Physiol* 2009

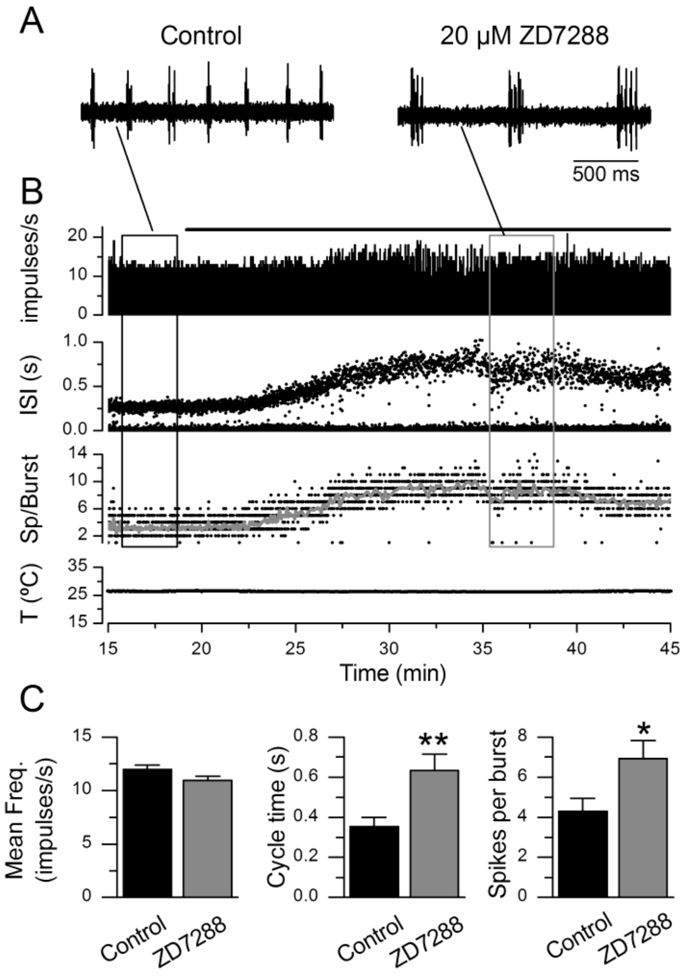
I_h blockade in CS nerve endings

Dramatic change in static firing pattern

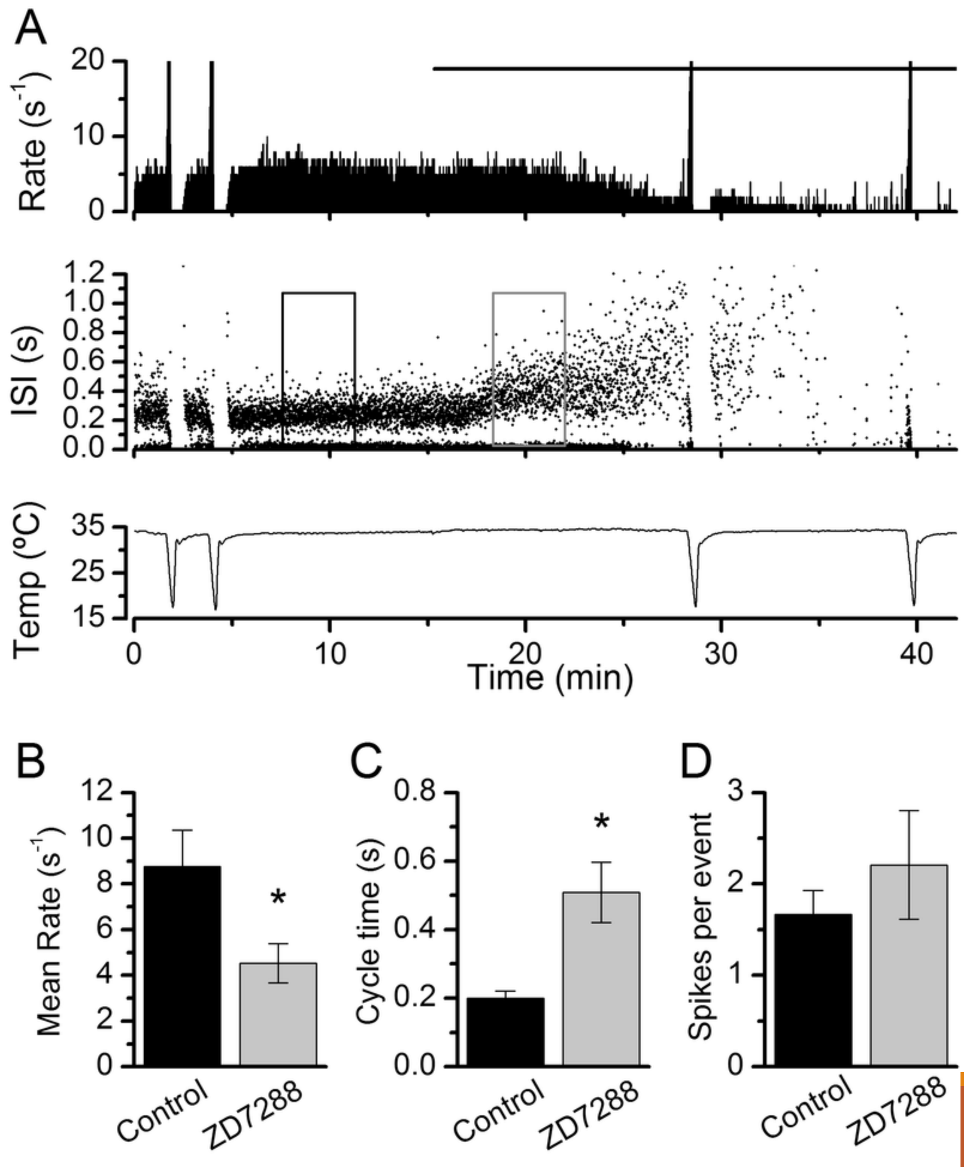
33ºC



26ºC



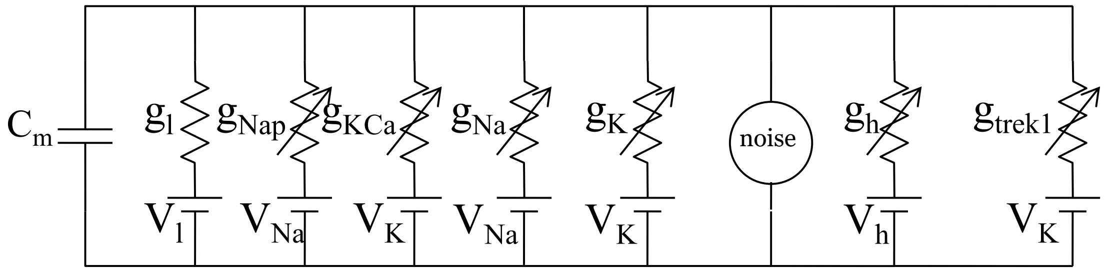
Effect of Ih blockade – mouse cornea

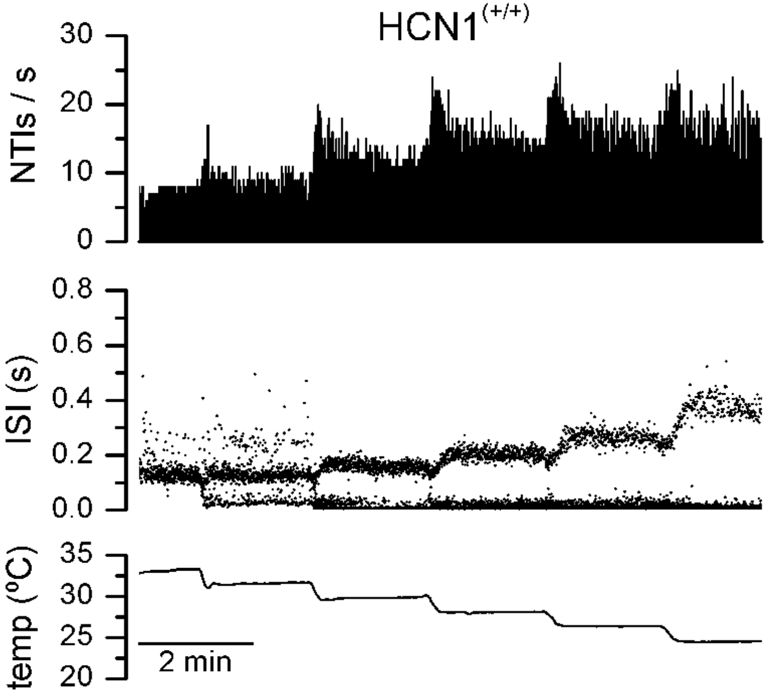


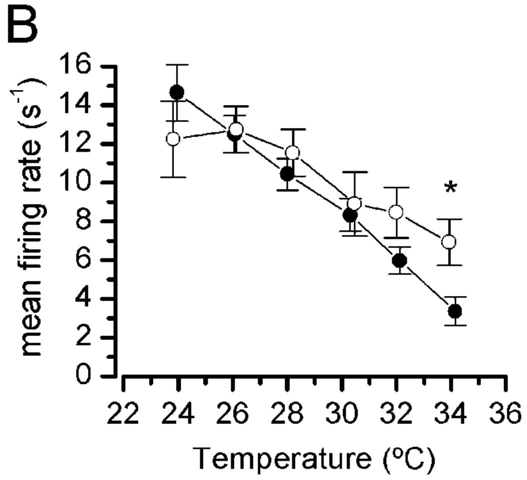
HCN1$^{(-/-)}$ mouse!!

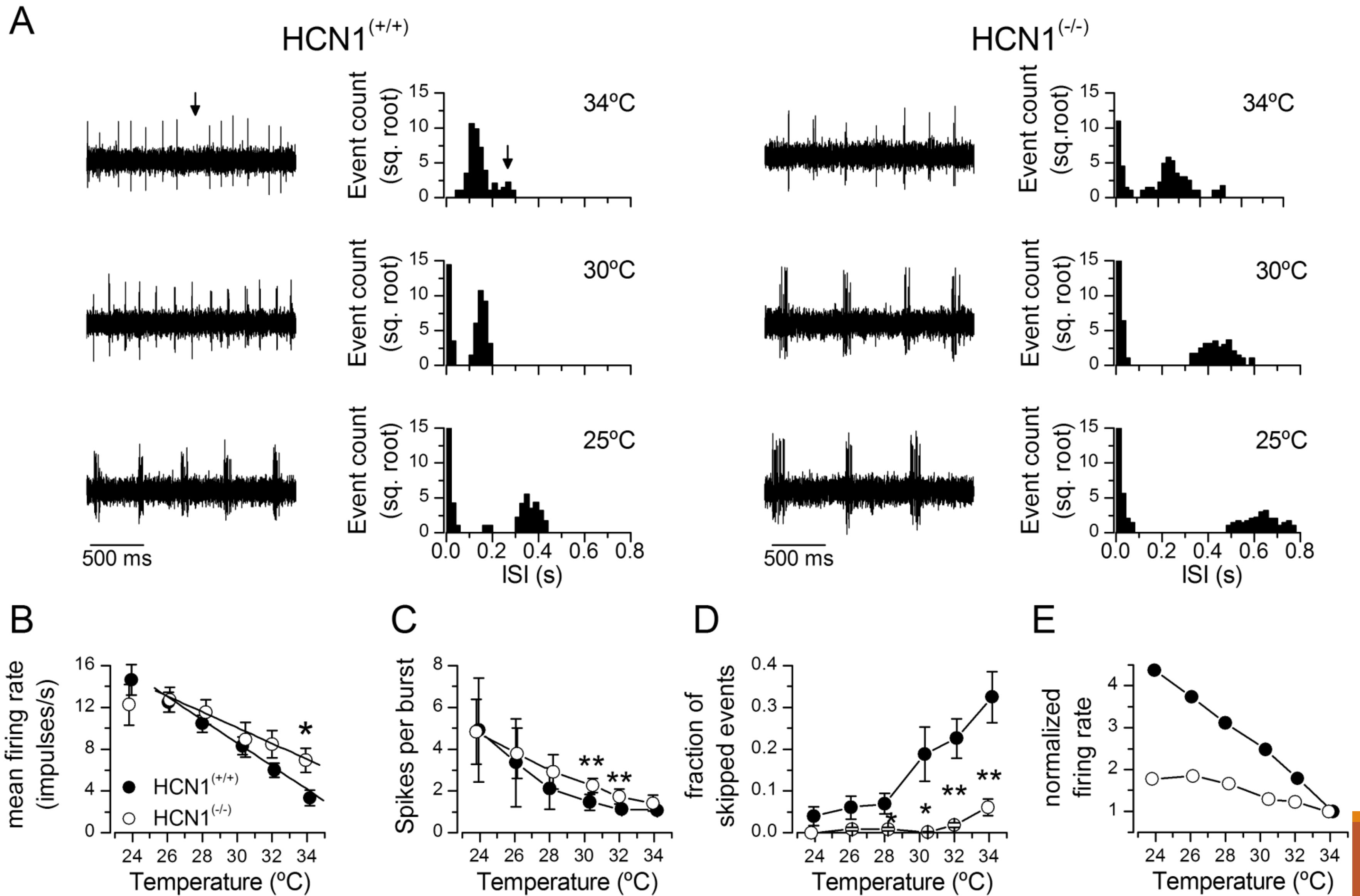
Can we explain the experimental results with a compensation by HCN2 in HCN1$^{(-/-)}$ animals?

Huber-Braun + Ih model

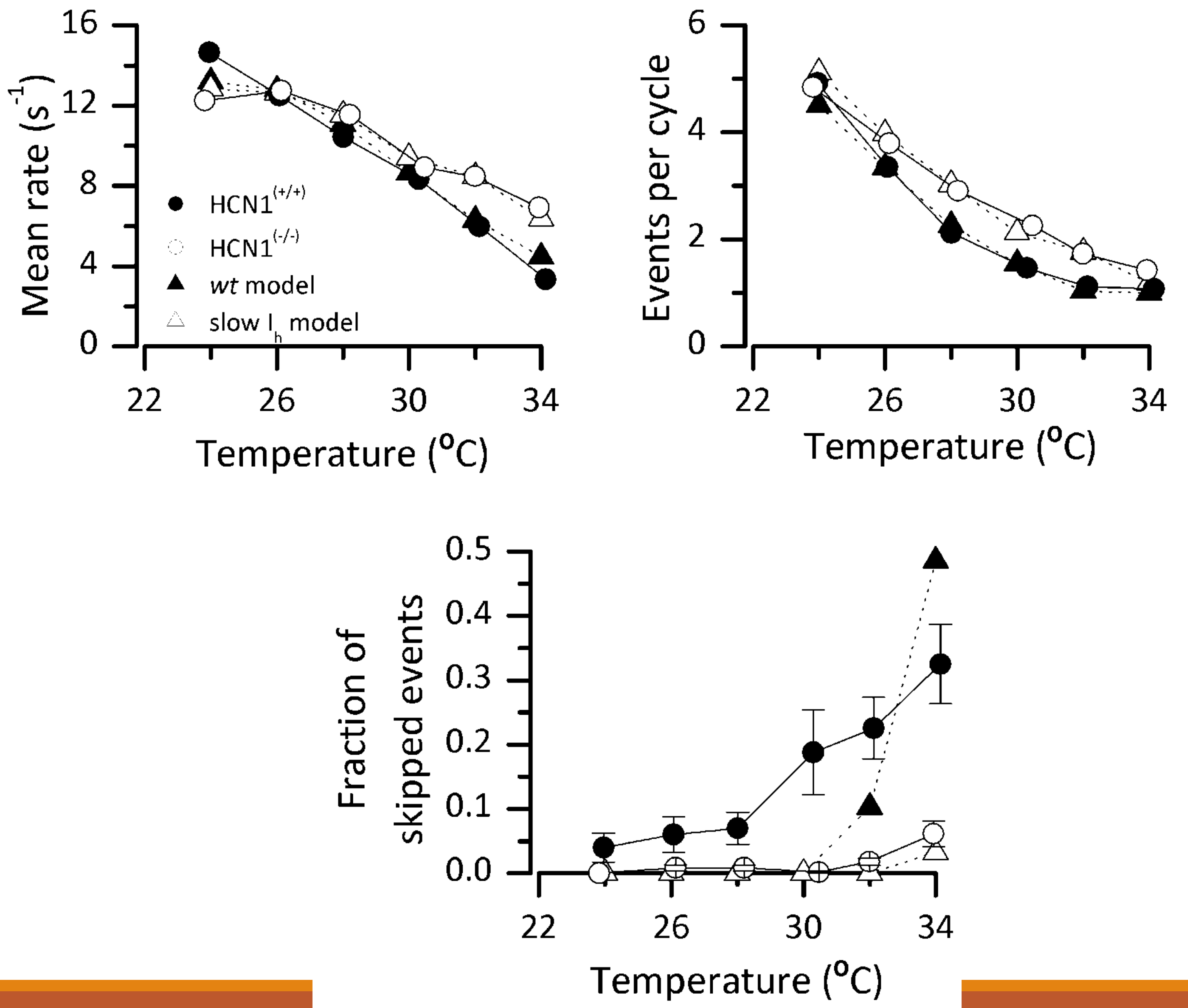




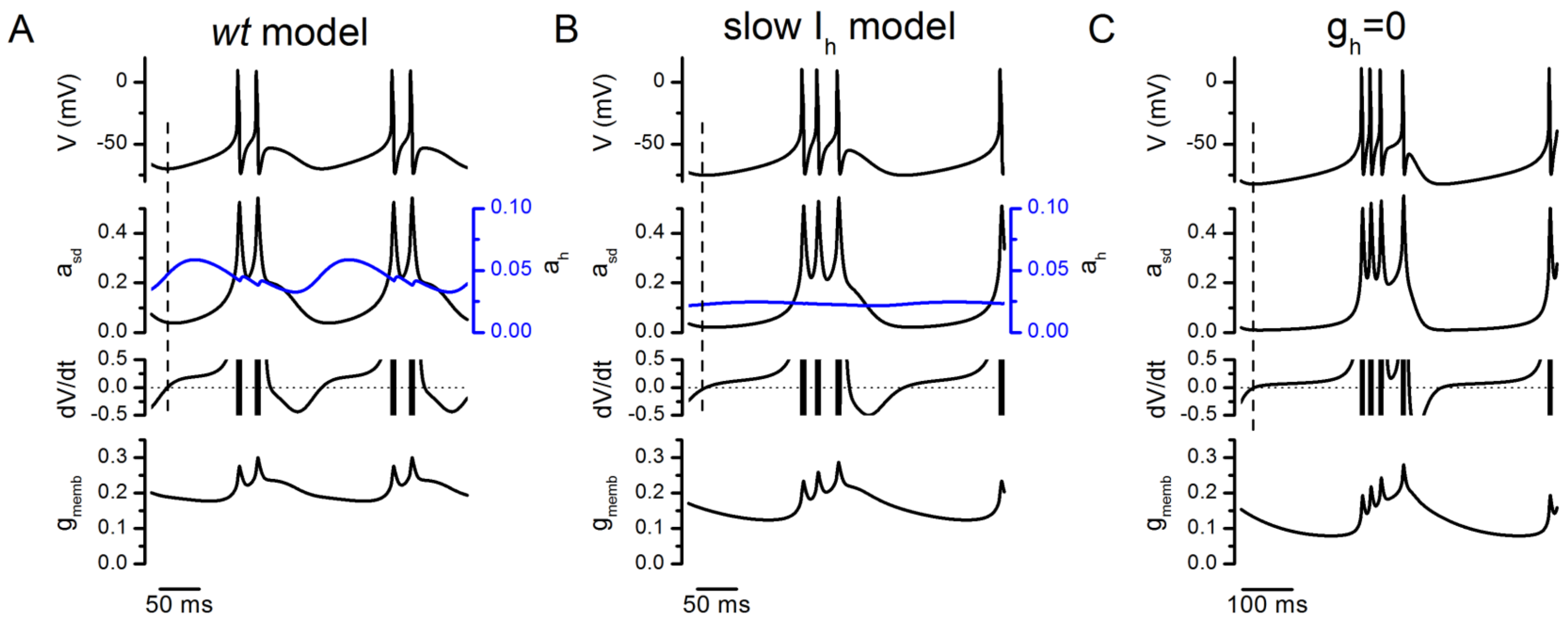




A
HCN1(+/+)
HCN1(-/-)
34ºC
30ºC
25ºC
Event count (sq. root)
Event count (sq.root)
500 ms
ISI (s)
B
mean firing rate (impulses/s)
HCN1(+/+)
HCN1(-/-)
Temperature (ºC)
C
Spikes per burst
Temperature (ºC)
D
fraction of skipped events
Temperature (ºC)
E
normalized firing rate
Temperature (ºC)


Mean rate (s^{-1})
16
12
8
4
0
HCN1(+/+)
HCN1(-/-)
wt model
slow I_h model
22
26
30
34
Temperature (°C)
Events per cycle
6
4
2
0
22
26
30
34
Temperature (°C)
Fraction of skipped events
0.5
0.4
0.3
0.2
0.1
0.0
22
26
30
34
Temperature (°C)

- I_h's feedback effect initiates the slow oscillations in CSNTs.
- Kinetics of HCN1 give the oscillations the right period.
- Experimental results obtained with HCN1$^{(-/-)}$ mice are better explained by a compensation with the related channel HCN2.
- Huber&Braun's model is expansible



Tools for modelers

- The Virtual Brain
  - https://www.thevirtualbrain.org/tvb/zwei/brainsimulator-software
- Brain Dynamics Toolbox
  - https://bdtoolbox.org
- BRIAN Simulator → spiking networks (and more)
  - https://briansimulator.org/
- NEURON → Detailed neurons, spiking neurons, networks.
  - www.neuron.yale.edu
- NETPYNE → NETworks in PYthon + NEuron
- DIY in Python or Matlab!!